

Credit Derivatives

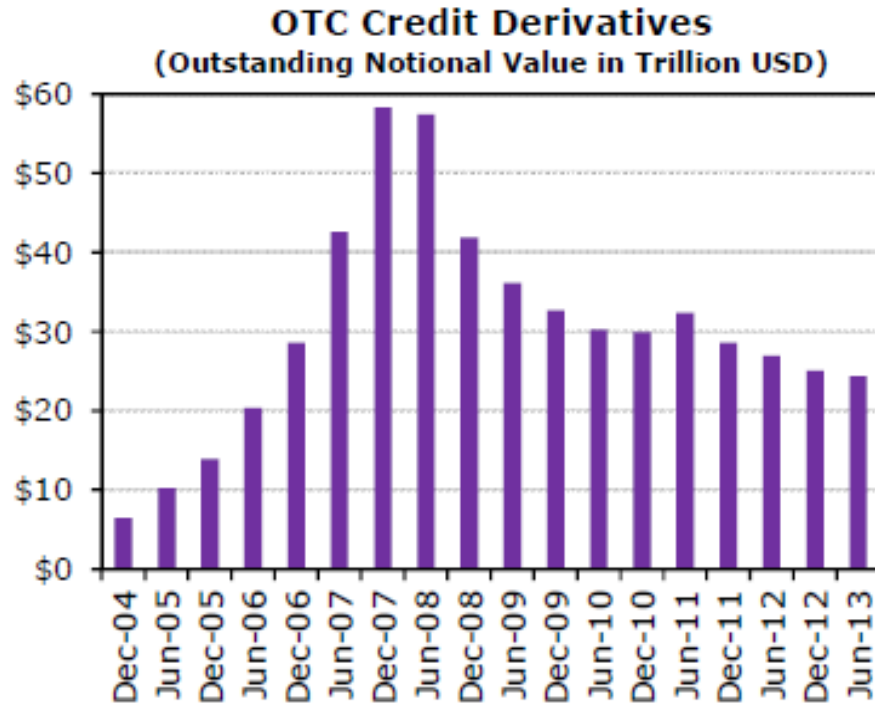
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February 2015

Overview

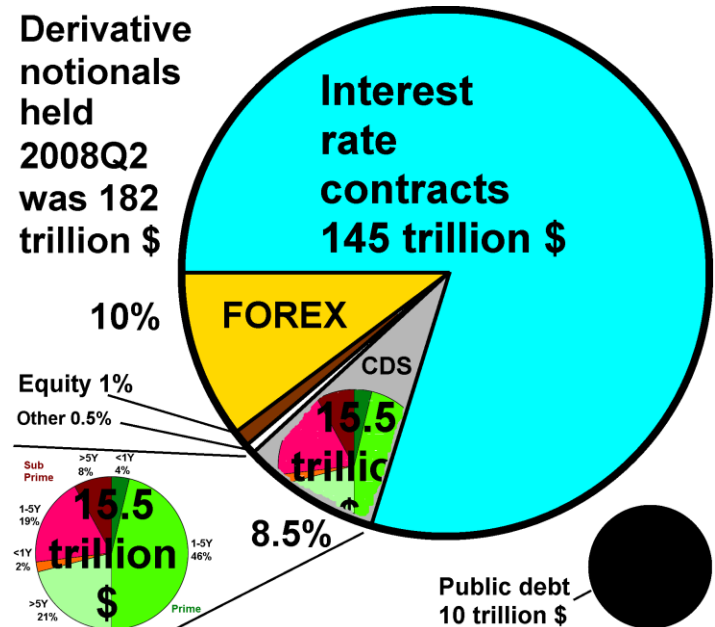
- Credit Default Swap
- Collateralized Debt Obligations
- Basel II Internal Ratings Based Approach

OTC Credit Derivatives Market Size



Source: Bank for International Settlements (BIS)

Derivative
notionals
held
2008Q2
was 182
trillion \$



CDS Definition

- In a standard ***credit default swap*** (CDS), a counterparty buys protection against default by a particular company or economic entity from a counterparty (seller).
- The company or entity is known as the ***reference entity*** and a default by that entity is known as a ***credit event***.
- The ***buyer*** of the CDS makes periodic payments or a ***premium*** to the ***seller*** until the end of the life of the CDS or until the credit event occurs.
- Depending on the contract, if the credit event occurs, the buyer has the right to sell a particular bond (or loan) issued by the company for its par value (physical delivery) or receive a cash settlement based on the difference between the defaulted bond's par value and its market price (recovery value) .

Credit Event

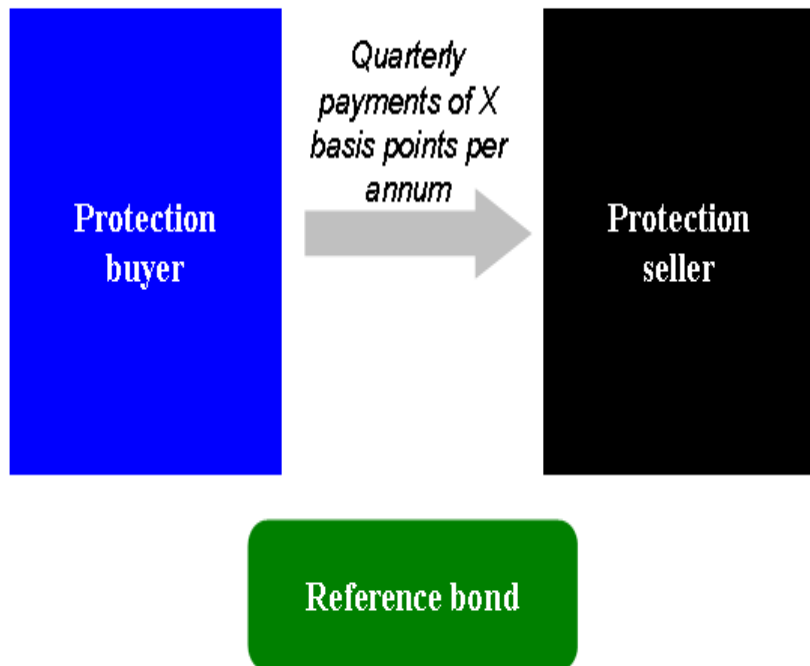
Declared by International Swaps and Derivatives Association (ISDA) Determination Committee.

Credit event usually assumes one of the following:

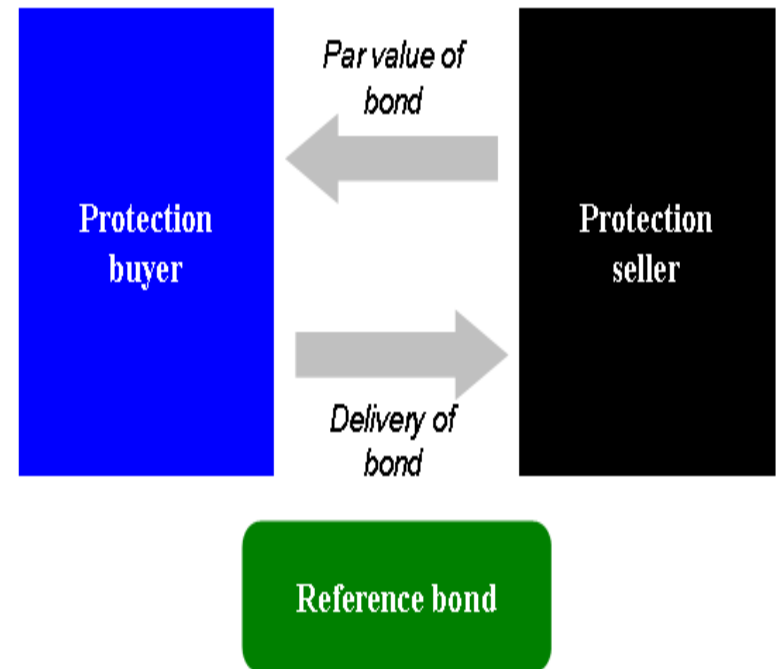
- Failure to meet payment obligations when due
- Bankruptcy
- Material adverse restructuring of debt
- Ratings deterioration

Credit Default Swap

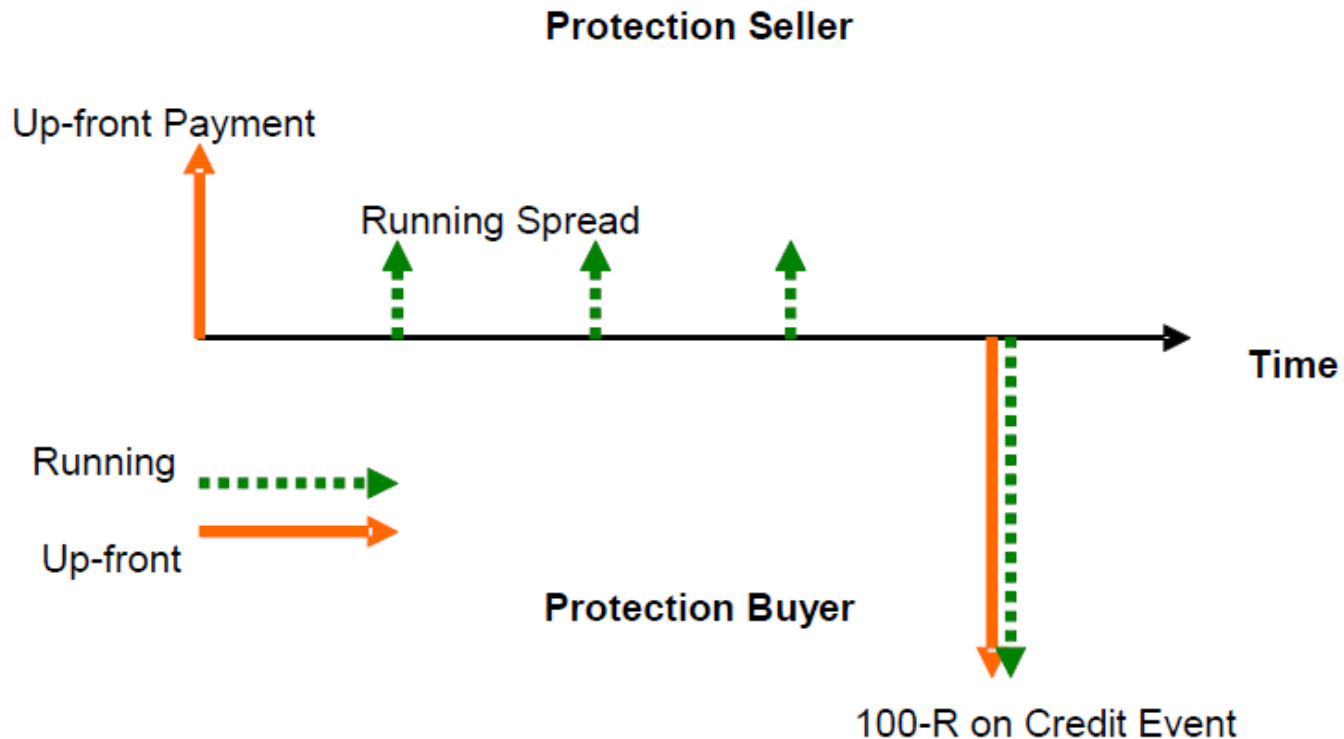
CDS (NO DEFAULT)



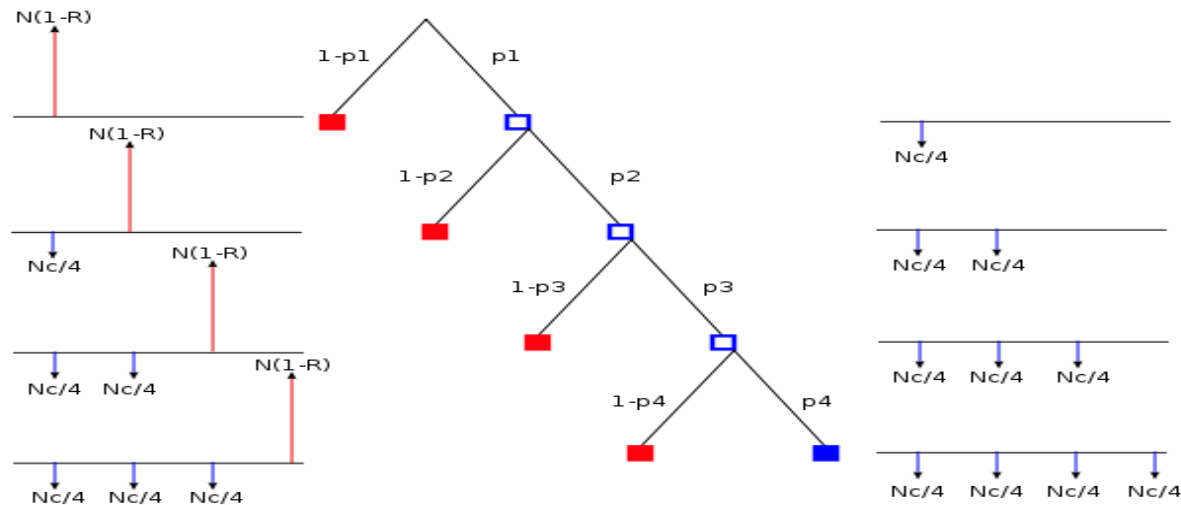
CDS (DEFAULT)



CDS Cash Flows



CDS Cash Flows



Description	Premium Payment PV	Default Payment PV	Probability
Default at time t_1	0	$N(1 - R)\delta_1$	$1 - p_1$
Default at time t_2	$-\frac{Nc}{4}\delta_1$	$N(1 - R)\delta_2$	$p_1(1 - p_2)$
Default at time t_3	$-\frac{Nc}{4}(\delta_1 + \delta_2)$	$N(1 - R)\delta_3$	$p_1p_2(1 - p_3)$
Default at time t_4	$-\frac{Nc}{4}(\delta_1 + \delta_2 + \delta_3)$	$N(1 - R)\delta_4$	$p_1p_2p_3(1 - p_4)$
No defaults	$-\frac{Nc}{4}(\delta_1 + \delta_2 + \delta_3 + \delta_4)$	0	$p_1 \times p_2 \times p_3 \times p_4$

Net Present Value of CDS Cash Flows

Premium leg:

$$PL(c) = N \cdot c \cdot \Delta t \left[p_1 (1 - p_2) D_1 + p_1 p_2 (1 - p_3) D_2 + p_1 p_2 p_3 (1 - p_4) D_3 + p_1 p_2 p_3 p_4 D_4 \right]$$

Default leg:

$$DL = N \cdot (1 - R) \cdot \left[(1 - p_1) D_1 + p_1 (1 - p_2) D_2 + p_1 p_2 (1 - p_3) D_3 + p_1 p_2 p_3 (1 - p_4) D_4 \right]$$

Net present value: $NPV = DL - PL(c)$

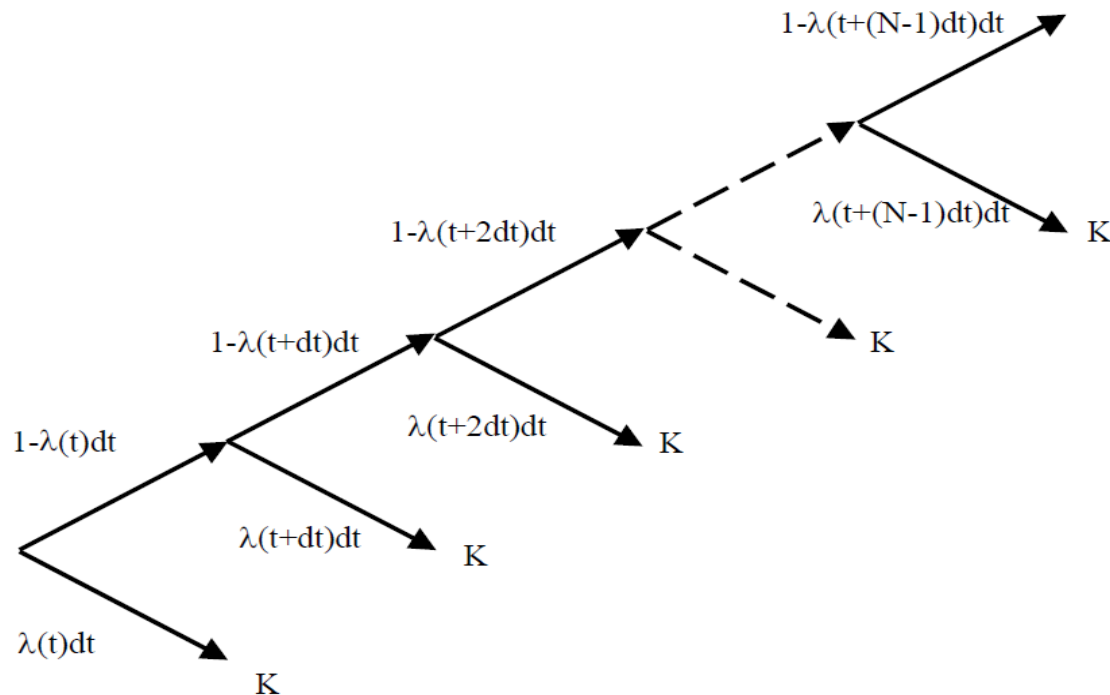
Breakeven spread: $NPV(s) = 0$

Reduced-Form Approach

Probability of default is described by a Poisson counting process:

$$P[\tau < t + dt \mid \tau \geq t] = \lambda(t) \cdot dt \rightarrow \text{Survival}(t) = \exp\left(-\int_0^t \lambda(u) \cdot du\right)$$

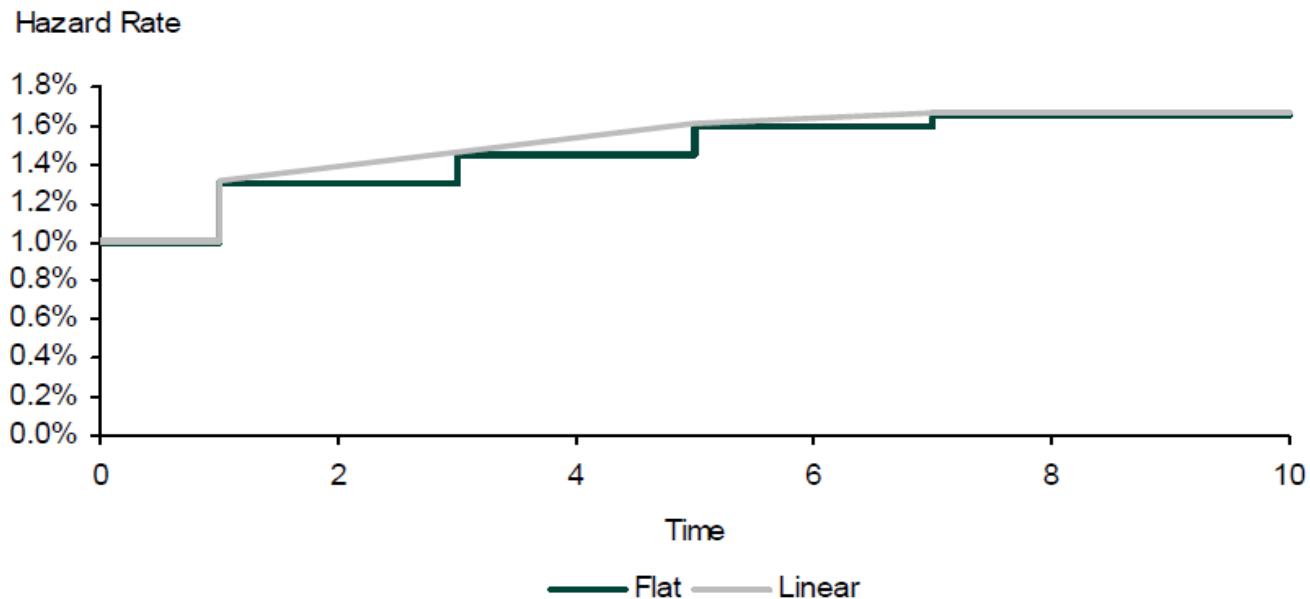
Default Process can be represented by a binomial tree in which tree terminates and makes a payment K at default



Reduced-Form Approach

Given the market default spreads at 1Y, 2Y, 3Y, 5Y and 10Y one can apply a bootstrapping procedure to find hazard rate term structure:

- Starting with the shortest maturity contract of 1Y calculate hazard rate for the first year
- Proceed with 2Y contract and calculate hazard rate for the second year
- Repeat until hazard rates for the longest contract are calculated

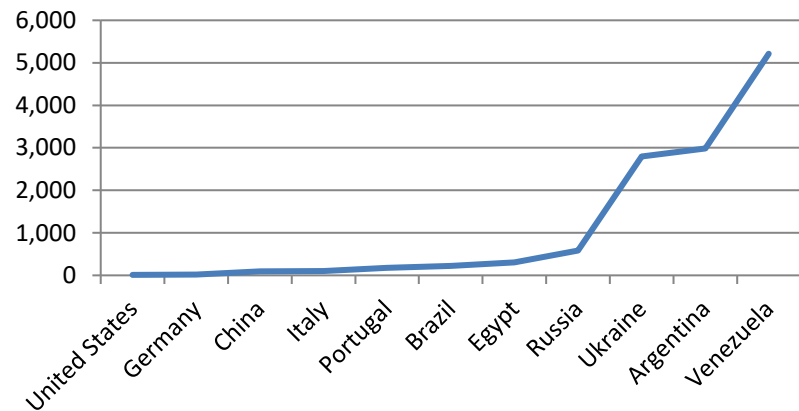


Sovereign Default Probabilities (from Deutsche Bank Research)

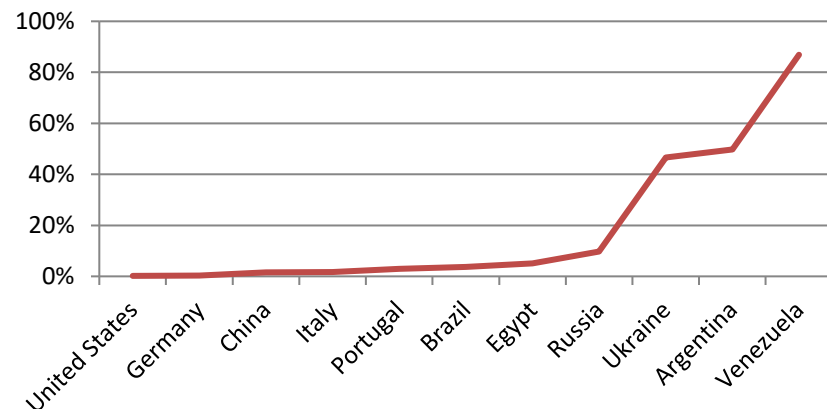
Country	1 Yr CDS Spread (Bp)	1 Yr PD
United States	17	0.3%
Germany	18	0.3%
China	93	2%
Italy	107	2%
Portugal	179	3%
Brazil	226	4%
Egypt	309	5%
Russia	583	10%
Ukraine	2,795	47%
Argentina	2,987	50%
Venezuela	5,209	87%

Approximation: $\text{Spread} = (1-R) * \text{PD}$

1 Yr CDS Spread (Bp)



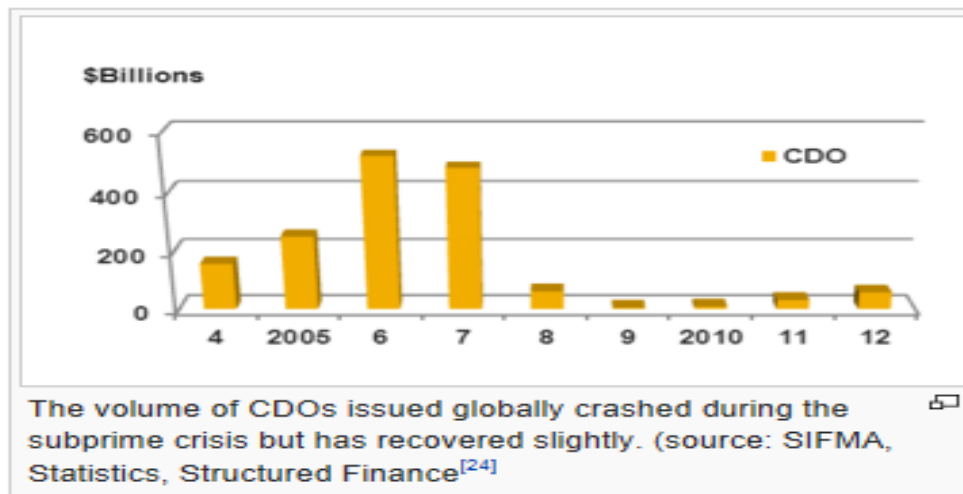
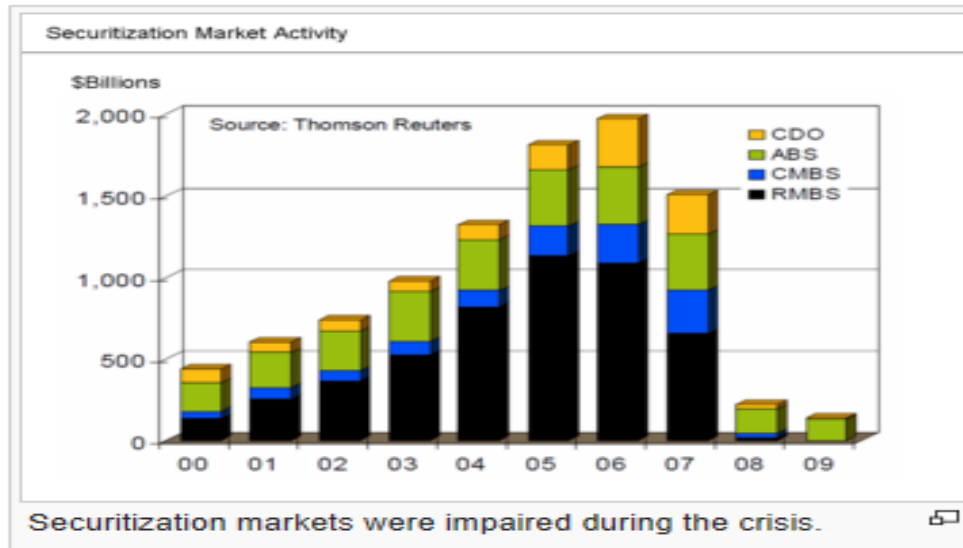
1 Yr PD (R=0.4)



Collateralized Debt Obligations

- CDO is a security whose income payments and hence value is derived from and collateralized by a specified pool of underlying assets.
- CDO is "sliced" into “tranches”, which "catch" the cash flow of interest and principal payments in sequence based on seniority.
- If some loans default and the cash collected by the CDO is insufficient to pay all of its investors, those in the lowest, most "junior" tranches suffer losses first.
- The last to lose payment from default are the safest, most senior tranches.
- Consequently coupon payments (and interest rates) vary by tranche with the safest/most senior tranches paying the lowest and the lowest tranches paying the highest rates to compensate for higher default risk.
- Synthetic CDO underlyings are comprised of CDS, and Cash CDO of loans.
- Separate special purpose entities—rather than the parent investment bank—issue the CDOs and pay interest to investors.

CDO Market Share



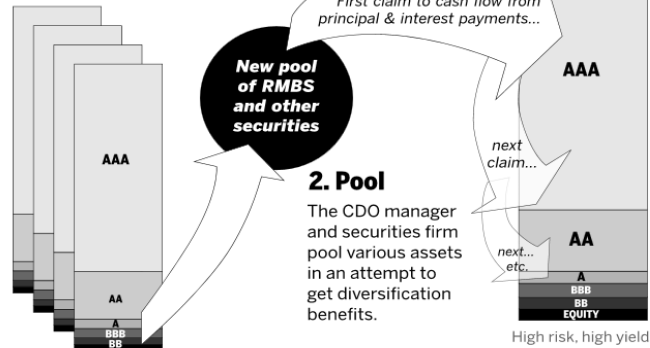
Structure of CDO

Collateralized Debt Obligations

Collateralized debt obligations (CDOs) are structured financial instruments that purchase and pool financial assets such as the riskier tranches of various mortgage-backed securities.

1. Purchase

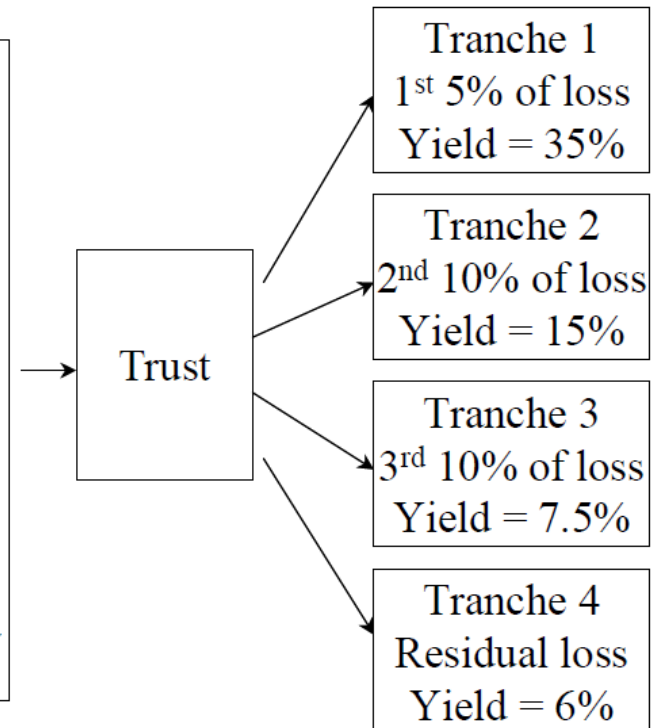
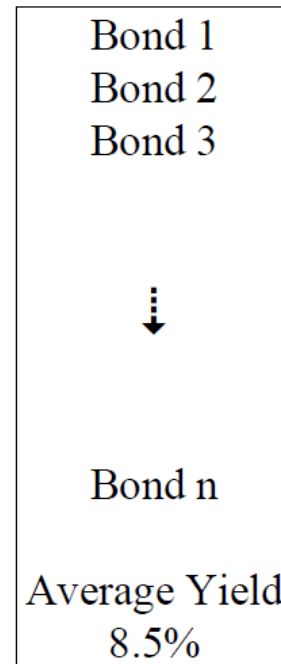
The CDO manager and securities firm select and purchase assets, such as some of the lower-rated tranches of mortgage-backed securities.



3. CDO tranches

Similar to mortgage-backed securities, the CDO issues securities in tranches that vary based on their place in the cash flow waterfall.

Low risk, low yield



iTraxx Index

- CDS index covers regions of Europe, Japan and non-Japan Asia.
- iTraxx Europe is composed of the most liquid 125 CDS referencing European investment grade credits.
- Constituents of this index are changed (rolled over) every six months on March 20th and September 20th.
- Index quotes are provided by Markit Ltd. (see)

Family	Type	Index name	Number of entities	Description
Europe	Benchmark Indices	iTraxx Europe	125	Most actively traded names in the six months prior to the index roll
		iTraxx Europe HiVol	30	Highest spread (riskiest) non-financial names from iTraxx Europe index
		iTraxx Europe Crossover	75	Sub-investment grade names
		iTraxx LEVX	40	European 1st Lien Loan CDS
		iTraxx Non-Financials	100	Non-financial names
		iTraxx Financials Senior	25	Senior subordination financial names
	Sector Indices	iTraxx Financials Sub	25	Junior subordination financial names
		iTraxx TMT	20	Telecommunications, media and technology
		iTraxx Industrials	20	Industrial names
		iTraxx Energy	20	Energy industry names
		iTraxx Consumers	30	Manufacturers of consumer products
		iTraxx Autos	10	Automobile industry names

Monthly iTraxx Tranche for March 28th, 2013

iTraxx Europe Series 15 Version 1 (RED Code: 2I666VAV7)
(<http://www.creditfixings.com/CreditEventAuctions/itraxx.jsp>)

Tranches 0-3, 3-6 are up-fronts with a fixed 500bps spread, Tranche 6-9 has a 300bps fixed spread. Tranches 9-12, 12-22, 22-100 are up-fronts with a fixed 100bps spread.

Tranche	5 Yr	Upfront (Bp)	Bid (Bp)	Mid (Bp)	Ask (Bp)	Dealers
Equity	0%-3%	500	3331	3481	3631	3
Junior Mezzanine	3%-6%	500	150	300	450	1*
Senior Mezzanine	6%-9%	300	-75	75	225	1*
Junior Senior	9%-12%	100	149	174	199	1*
Senior	12%-22%	100	63	80.5	98	1*
Super Senior	22%-100%	100	15	17.5	20	1*

CDX Index

- CDS index covers regions of North America and Emerging Markets.
- CDX.NA.IG is composed of equally weighted 125 CDS investment grade CDSs.
- A new series of CDS indices is issued every six months by Markit Group Limited. This process is intended to ensure that the index does not become "cluttered" with instruments that no longer exist, or which trade extremely with low liquidity.

Index Name	Number of entities	Description
CDX.NA.IG	125	Investment grade CDSs
CDX.NA.IG.HVOL	30	High Volatility investment grade CDSs
CDX.NA.HY	100	High Yield CDSs
CDX.NA.HY.BB	37	Index of high yield CDSs with a BB rating
CDX.NA.HY.B	46	Index of high yield CDSs with a B rating
CDX.NA.XO	35	CDSs that are at the crossover point between investment grade and junk
CDX.EM	14	Emerging market CDSs
CDX.EM Diversified	40	Emerging market CDSs
LCDX	100	NA First Lien Leverage Loans CDSs

Pricing CDO Tranche

Premium leg: $PL = S \cdot \sum_{i=1}^n \Delta t_i \left[1 - EL_{(K_1, K_2)}(t_i) \right] \cdot D(t_0, t_i)$

Discount factor: $D(t_0, t_i) = \exp(-r(t_i - t_0))$

Default leg: $DL = \sum_{i=1}^n \left[EL_{(K_1, K_2)}(t_i) - EL_{(K_1, K_2)}(t_{i-1}) \right] \cdot D(t_0, t_i)$

Expected loss: $EL_{(K_1, K_2)} = \frac{1}{K_2 - K_1} \sum_{i=1}^n (\min(L_i, K_2) - K_1)^+ \cdot p_i$

Given portfolio cumulative loss distribution function F ,

$$EL_{(K_1, K_2)} = \frac{1}{K_2 - K_1} \left(\int_{K_1}^1 (x - K_1) dF(x) - \int_{K_2}^1 (x - K_2) dF(x) \right)$$

One Factor Model

We assume that portfolio of referenced assets consists of N financial instruments. Asset return is described by common and idiosyncratic risk factors:

$$A_i = \rho M + \sqrt{1 - \rho^2} X_i$$

Assume that default occurs when the asset return crosses the threshold C_i , which is implied by obligor's default probability:

$$PD_i = P[A_i \leq C_i] = \Phi(C_i) \rightarrow C_i = \Phi^{-1}(PD_i)$$

Issuer defaults if, $X_i \leq \frac{C_i - \rho M}{\sqrt{1 - \rho^2}}$

Probability of default conditional on common factor M is,

$$p_i(M) = \Phi\left(\frac{\Phi^{-1}(PD_i) - \rho M}{\sqrt{1 - \rho^2}}\right)$$

One Factor Model

Probability that k out of N issuers default is,

$$P[L = L_k | M] = \binom{N}{k} \cdot p^k(M) \cdot (1 - p(M))^{N-k}$$

Cumulative probability of percentage loss net exceeding θ is,

$$F_N(\theta) = \sum_{k=0}^{[N\theta]} \int_{-\infty}^{\infty} \binom{N}{k} \cdot p^k(u) \cdot (1 - p(u))^{N-k} d\Phi(u)$$

Cumulative distribution of losses for a large homogeneous portfolio is,

$$F_{\infty}(\theta) = \Phi\left(\frac{\sqrt{1 - \rho^2} \Phi^{-1}(\theta) - \Phi^{-1}(PD)}{\rho}\right)$$

Gaussian Distribution

Plugging in Gaussian distribution in equation for tranche losses, we get:

$$EL_{(K_1, K_2)} = \frac{(1-R)}{K_1 - K_2} \cdot \left[\Phi_2 \left(-\Phi^{-1} \left(\frac{K_1}{1-R} \right), \Phi^{-1}(PD); \Omega \right) - \Phi_2 \left(-\Phi^{-1} \left(\frac{K_2}{1-R} \right), \Phi^{-1}(PD); \Omega \right) \right]$$

Here Φ is the cumulative normal distribution, Φ_2 is the cumulative bi-variate normal distribution, and Ω is the covariance matrix:

$$\Omega = \begin{pmatrix} 1 & -\sqrt{1-\rho^2} \\ -\sqrt{1-\rho^2} & 1 \end{pmatrix}$$

Normal-Inverse Gaussian Distribution

NIG distribution is fat-tailed and skewed. Its probability density function is,

$$f_{NIG}(x; \alpha, \beta, \mu, \delta) = \frac{\delta \alpha \cdot \exp(\delta \gamma + \beta(x - \mu))}{\pi \sqrt{\delta^2 + (x - \mu)^2}} \cdot K_1\left(\alpha \sqrt{\delta^2 + (x - \mu)^2}\right)$$

Here $\gamma = \sqrt{\alpha^2 - \beta^2}$, and K_1 is the modified Bessel function of the third kind,

$$K_1(\omega) = \frac{1}{2} \int_0^{\infty} \exp\left(-\frac{\omega}{2}(t + t^{-1})\right) dt$$

Parameters: μ - location, α - tail heaviness, β - asymmetry, δ - scale

Moments: $Mean = \mu + \delta \beta / \gamma = 0$ $Variance = \delta \alpha^2 / \gamma^3 = 1$

$$Skewness = 3\beta / (\alpha \sqrt{\delta \gamma}) \quad Kurtosis = 3(1 + 4\beta^2 / \alpha^2) / (\delta \gamma)$$

Monte Carlo Algorithm

- Find Cholesky decomposition A of correlation matrix $R=AA^*$.
- Simulate n independent random variables $Z = (Z_1, Z_2, \dots, Z_n)'$ from normal distribution, where n is the number of paths.
- Let $Y=AZ$, then Y follows multivariate normal distribution with correlation R .
- Obtain default times $(T_1, T_2, \dots, T_n)$ by mapping back the Y 's to the T 's by using equation, $\Phi(Y) = 1 - \exp(-\lambda t)$ for default probability, where Φ is the cumulative normal distribution function.
- Select subset of default times that are smaller than simulation time t .
- Compute cumulative loss of portfolio of M names, $L(t) = N(1-R) \sum_{i=1}^M 1_{\{t_i < t\}}$
- Compute each tranches loss, default and premium legs.
- Compute breakeven spreads.

Bank Capital Regulation

(Basel II Internal Ratings Based Approach)

α - Percentile of the loss fraction distribution of a Large Homogenous Portfolio:

$$\alpha = \Phi \left[\frac{\sqrt{1-\rho^2} \Phi^{-1}(x_\alpha) - \Phi^{-1}(PD)}{\rho} \right] \rightarrow x_\alpha = \Phi \left[\frac{\Phi^{-1}(PD) + \rho \cdot \Phi^{-1}(\alpha)}{\sqrt{1-\rho^2}} \right]$$

Capital requirement:

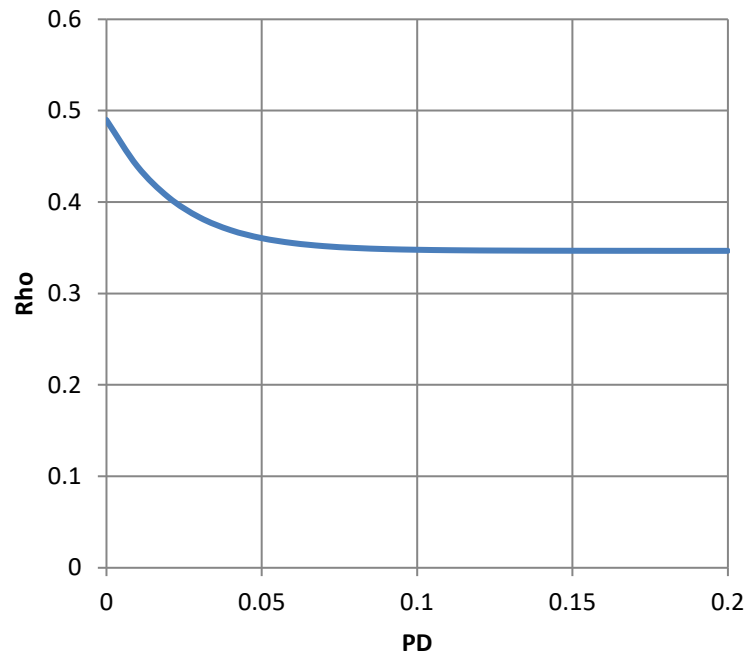
$$K = (1-R) \cdot x_{0.999} = (1-R) \cdot \Phi \left[\frac{\Phi^{-1}(PD) + \rho \cdot \Phi^{-1}(0.999)}{\sqrt{1-\rho^2}} \right]; RWA = 12.5 \cdot K \cdot EAD$$

Correlation:

$$\rho = \sqrt{0.24 - 0.12 \cdot \frac{1 - \exp(-50 \cdot PD)}{1 - \exp(-50)}}$$

Bank Capital Regulation

Correlation



Capital Requirement (R=0.4)

